

[5460]-190

T.E.(Computer Engineering)

DIGITAL SIGNAL PROCESSING APPLICATIONS

(2012 Pattern)

Time : 3 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Assume suitable data if necessary.

- Q1)** a) How the linear convolution operation can be used to represent any arbitrary DT sequence and a DT system? [5]
 b) Obtain the Z — Transform (ZT) of a DT signal - [5]
 $x(n) = -a^n u(-n-1)$ Sketch the ROC.

OR

- Q2)** a) A CT signal having frequency 50 Hz is sampled at a rate of 1200 samples/sec. Obtain - [5]
 i) Number of samples per cycle.
 ii) Digital/Discrete frequency f and ω .
 iii) Minimum sampling rate to avoid aliasing effect.
 iv) Period of a DT signal.
 b) State and prove the periodicity property of Fourier Transform (FT). Define it for DFT. [5]

- Q3)** a) What is the convolution property of DFT? Compare Linear Convolution with Circular Convolution. [5]
 b) Derive the first stage of Radix-2 DIT FFT Algorithm. [5]

OR

- Q4)** a) State the relation between FT and DFT. Define N point DFT by means of twiddle factor and obtain the twiddle factors for 4 point DFT. [5]
 b) Obtain the system function and impulse response of the given system described as- [5]

$$y(n) - \frac{5}{6}y(n-1) + \frac{1}{6}y(n-2) = x(n) - \frac{1}{2}x(n-1)$$

P.T.O.

- Q5)** a) Derive the Direct Form - I IIR filter structure from the system function $H(Z)$ and realize it using multipliers, adders and delay elements. [9]
 b) Obtain and realize linear phase FIR filter structure having impulse response. [9]

$$h(n) = \delta(n) + \frac{1}{2}\delta(n-1) - \frac{1}{4}\delta(n-2) + \frac{1}{2}\delta(n-3) + \delta(n-4)$$

OR

- Q6)** a) Obtain and draw the cascade form realization for IIR filter having transfer function [9]

$$H(Z) = \frac{Z^2 - Z}{Z^2 - 0.2Z - 0.15}$$

- b) Represent the mathematical form of M^{th} order FIR filters by means of system function $H(Z)$. Draw the Direct Form filter structure and determine the number of multipliers, adders and delay elements required to realize the filter. [9]
- Q7)** a) Explain the features of SHARC DSP processor. List the number of DAGs with its capabilities and memory pointer registers supported by DAG. [8]
 b) What is OMAP? Explain the features and applications of OMAP in brief. [8]

OR

- Q8)** a) Compare conventional Microprocessor with DSP Processor architecture. Draw and explain basic building blocks of DSP processor. [8]
 b) Draw and explain the SIMD (Single Instruction Multiple Data) architecture of SHARC DSP processor. [8]

- Q9)** a) Draw and explain Human Speech Model in speech synthesis and recognition. [8]
 b) How digital image is represented by means of digital computer? How gray scale image is different than colour image? What is Histogram of an image? [8]

OR

- Q10)** a) What is Companding? What is its significance in audio processing? What is the impact of data rate on sound quality? [8]
 b) With mathematical form, explain any two gray level transforms used for image enhancement. [8]

